INTERNTRACK: INTERNSHIP JOURNAL AND PROGRESS MONITORING SYSTEM

GROUP 1

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CHAPTER I

Introduction

1.1 Project Context

Technology has enhanced every aspect of education in today's learning institutions, inside and outside the classroom. These days, colleges need readily available functional technology. It is more of a practical need than a want. One way technology has been beneficial in education is through well-organized programs that allow students to experience what they have learned in school and apply it in the real world.

Mater Dei College - Cabulijan, Tubigon, Bohol, requires all its students to accomplish an internship, also known as OJT. Internships are part of the course requirement for all tracks, including Information Technology, Business Administration, Accounting, Hospitality Management, and Tourism. While inside assigned to their company, students should keep a daily log of their activities, be evaluated by their assigned company, and be monitored by their coordinators.

Currently, the college does not have a centralized mechanism for submitting OJT journals. As a result, data is processed differently across departments, compliance is not easily monitored in real-time, and students and coordinators must spend time entering and reviewing periodic statistics. Additionally, partner company supervisors do not have a formal process for reviewing, approving, or accepting such logs submitted by their interns.

InternTrack: A Web-Based Internship Journal and Progress Monitoring System was developed to address the gaps in implementing the policy and technology and digitize the entire OJT journal – keeping process into a more transparent, systematic, and accessible system for students, company supervisors, and academic program coordinators.

1.2 Purpose and Description

InterTrack was created for Mater Dei College to organize and provide online management for the documentation of the OJT journal module during their undergraduate courses. This maintains four user types varying levels: students, company supervisors, academic coordinators, and system administrators. Each level grants different user rights based on predefined roles.

Through the system, students can write, view, and manage their daily journal, monitor its submission status through an interactive calendar, and download PDF copies of their journal entries for the entire internship period. Students are notified by email, on the web, and via mobile applications about overdue and missing journal entries. Students can also see a list of overdue, missing journal entries. Company supervisors also have another window to view and comment on student's weekly journals from other batches before approving or sending them back. Administrators handle account registrations, batch configurations, and platform settings. In addition to journal and workflow management, the system also digitally manages the key SIPP compliance documentation required by the college. These are the Student Information Sheet, and the Annual Report on the Implementation of the Student Internship Program in the Philippines (SIPP). When the student information is submitted, it automatically enters into the coordinator's Student Information Sheet. The system allows for the creation of Host Training Establishments (HTEs) compliant with SIPP requirements and generates Annual SIPP Reports. All of these can be encoded, stored, and generated within the platform by authorized users.

In addition to individual functionality, InternTrack features weekly automated bundling of daily entries into a weekly log on Sundays, as well as an automatic warning that notifies students of incomplete or missing entries via their school emails and via push notifications on the mobile app. The implementation of InternTrack aims to reduce journal documentation time, enhance school oversight, and lighten the faculty coordinator's load throughout the OJT period.

1.3 Objectives

General Objective:

The main goal of this project is to create, program, and launch a responsive web-based and mobile application OJT Journal and Progress Monitoring System specifically for Mater Dei College. This system will cater to internship programs related to the fields of Information Technology, Business Administration, Accounting, Hospitality Management, Tourism by allowing students to submit daily journal entries easily, utilize the OJT journal's built-in review and approval process that consists of many tiers, such as immediate supervisors and upper management coordinators, and benefit from the journal's many report generation tools to monitor student progress easily.

Specific Objectives:

- To design a student-centered web portal that enables interns to write, view, edit, and organize journal entries related to their daily tasks.
- To implement a process that compiles daily entries into a single weekly file each Sunday for supervisor approval, and afterward transfers approved files to the coordinator.
- To integrate a calendar view that will provide students with a quick status check for daily entries, completed, missing, or overdue submissions.
- To build a supervisor dashboard where they can review, write feedback, and accept or return weekly journals for student editing. Ensure an easily accessible log of edit history is available throughout the review process.
- To develop a scalable report engine that enables academic coordinators and students to generate structured, date-restricted PDF documents containing all validated journal entries.
- To include an auto-email reminder script that emails students about missing or late daily logs at 9:00 PM, customized by coordinators for different batches of students, and sends these reminders to the mobile app as push notifications.

- To develop a secure administrator panel to handle user accounts, student batches, and customize systemwide default settings.
- To deliver a responsive web application that operated effectively across desktop and mobile browsers, enabling reliable and consistent access for college coordinators, company supervisors, and students.
- To allow authorized users to generate, manage, and encode SIPP compliance documents within the system, such as the Student Information Sheet, and the Annual SIPP Implementation Report, to meet regulatory and institutional reporting needs.
- To implement Weekly Activity Log functionality for students to record and report highlights of their weekly activities, as well as automatically include approved Journal Entry input for the Student SIPP report available to coordinators.
- To provide the coordinator's overdue and missing entries navigation panel with lists of all interns who failed to submit their entries, enabling easy tracking and sending notices to students.
- To automate the Student Information Sheet page that will populate for the coordinator upon students' info submission, and add Partner Companies along with Head of Company, Department Head, and Supervisor hierarchies, inclusive of contact information.
- To support HTEs in meeting SIPP requirements and the creation of annual reports detailing the Student Internship Program in the Philippines (SIPP) implementation.

1.4 Scope and Limitations

Scope:

The InternTrack system is intended to provide a functional and structured web-based environment for managing the key workflows of the OJT journaling and monitoring process. The system covers the following areas:

- Journal Management: Students may upload and edit their daily journals with provided headings, sections, and checkboxes, rather than a rigid template. The student can still modify submitted journals until they are workflow-locked.
- Calendar Overview: Displays a calendar with daily journal statuses color-coded. Status depends on the number of working days set against the batch.
- Weekly Processing: The system automatically collects daily logs into a weekly format every Sunday evening, including only entries supplied inside the active batch schedule.
- Supervisor Workflows: Allows supervisors to enter written comments when approving or returning a student’s weekly log within this environment. Students can edit and resubmit their logs until they receive approval. Approved records will be forwarded to a college coordinator.
- Coordinator Tools: A portal for administering programs and tracking multiple student groups. Includes read-only access to all records approved by supervisors.

- Report Generation: Journal details for a specified date range can be readily viewed and printed by students and coordinators. Users can generate comprehensive reports and download them in PDF format for documentation and submission purposes.
- Notification Engine: Send automatic email and mobile push notifications to students with open journals who have not submitted partially complete, based on reading batch parameters.
- Account and System Setup: Allow registered Administrators to create new users, assign roles, create batches, and configure global system settings.
- Profile Storage and Compliance Documents: A unique platform to digitally encode, store, and retrieve key SIPP compliance documents, including the Student Information Sheet, and Annual SIPP Report.
- Weekly Activity Log: Students can record and submit their weekly activities. Approved daily logs will be automatically reflected in their SIPP report, which coordinators can view.
- Overdue and Missing Tracking: Coordinators will have a navigation panel that lists all interns with pending daily journals. Coordinators can view details and notify students directly from this panel.
- Partner Company: The system supports a company hierarchy comprising the Head of Company, Department Head, and Supervisor. Also includes company contact numbers and information for each partner company.

- Student Information Sheet – Students and coordinators can view it. When the student submits their information, it will automatically be entered into the coordinator’s Student Information Sheet.
- HTE and Annual Report (SIPP): The system will allow coordinators to set up Host Training Establishments (HTE) as required by SIPP. Coordinators will also be able to create an Annual Report on the implementation of the SIPP.

Limitations:

To maintain a clear and realistic project scope, the system operates within the following constraints:

- Department Coverage: The program was created specifically for Information Technology, Business Administration, Accounting, Hospitality Management, and Tourism students at Mater Dei College. It is not suitable for departments at Mater Dei College that are not involved in these programs to use this system for their OJT programs.
- Hour Tracking: This application serves only as a platform for documenting weekly journal entries and for verification. There is no tracking, accumulating, or entry of total internship hours through this app. Hours are kept completely separate.
- Connectivity Requirement: The platform needs a stable internet connection for users to access dashboards, save their work, and submit reports. However, Offline drafts and local saving are features now available but

require internet connection to use, so users should make sure they are connected to the internet before using the platform.

- Post-Deadline Submission Policy: Entries marked as overdue at the time of report compilation will be excluded from the weekly report, even if additional late submissions occur.

CHAPTER II

Review of Related Literature

The Interntrack: Internship Journal and Progress Monitoring System was created by the developers to help improve how internships are managed. Today, many educational institutions use digital systems to make academic processes more efficient and to connect students, teachers, and administrators. This project reviews different sources to understand how technology can support monitoring and accountability in internships. The literature review discusses key ideas and principles, including internship documentation, progress reporting, and ways to improve the system.

Evaluation of an Online Internship Journal System for Interns – conducted by Lih-Juan ChanLin, is among the notable studies on technology-supported internship documentation in higher education. The researchers wanted to see how online journal writing could help students reflect on their internship experiences. They developed an online journal system to manage content for Fu Jen Catholic University's Department of Library and Information Science. They offered professional skill training with support from digital content management. The system consisted of a content management system, a 23-question learning journal, and features of the system and management. The study focused on reflective writing, critical thinking, and building competence. Data from 70 interns showed that most students had a positive experience with the learning and reflection involved in journal writing.

The On-The-Job Training Experiences of Business Administration Students of Philippine Electronics and Communication Institute of Technology in Butuan City – created by Hingpit S. (2024). In this study, students experienced problems with mismatched skills on assigned tasks, inadequate supervision, and a lack of monitoring by advisers and supervisors. The researcher highlighted the importance of implementing policies and a monitored structure to maintain effective OJT programs.

Web-Based Internship Information System – developed by N. Hasti and S. Lestari (2019) - is among the notable digital solutions for streamlining and managing the internship process in educational institutions. The researchers developed this system to focus on how web-based technology can solve problems at every stage, from registration to appraisal. Built using an object-oriented approach and the Prototype Method, the system includes features such as registration, report submission, and appraisals. It focuses on accuracy, efficiency, and quick information delivery. The results showed that the system reduced errors and gave students timely information about their internships.

Design and Implementation of Web-Based Internship Information System at Vocational School – developed by Yannuar, Hasan, Gafar Abdullah, Hakim, and Wahyudin (2018)- aims to improve how vocational schools manage internships, especially in areas with low technology use. The researchers saw the need for a digital platform to connect students, supervisors, industry partners, and coordinators. The system was built using PHP, Bootstrap, CSS, HTML, JavaScript, and MySQL. It includes features such as daily journals, portfolio and final reports, scheduling, and the organized documentation of internship activities. The study found that the system helped all parties manage internships more efficiently and accessibly.

Of the presented studies and systems, none yet addresses the need for a comprehensive web-based system that supports journal documentation, workflow management, and the dissemination of internship company information in a single, organized package tailored to the setting and needs of Mater Dei College. The following systems explored technologies for online journaling, monitoring and approval processes, and the internship information delivery; however, they were either limited to covering each internship phase individually or were not designed for academic institution in the Philippines. InternTrack differs from these previously studied systems in that it addresses each gap mentioned in this literature review through a single end-to-end platform complete with multi-level journal review and approval process workflow engines, SIPP compliance documents, and mobile application support for students, company supervisors, academic coordinators, and administrators.

CHAPTER III

Technical Background

This chapter describes the project from a technical perspective by explaining its fundamental operating principles, the development technologies employed, and its key functionalities.

3.1 Architecture Design

The group plans to create a web application that students, company supervisors, academic coordinators, and system administrators can use on any device with internet access. With this system, users will be able to keep journals, monitor their submissions, check calendars, and approve weekly reports. It will also help make account management easier for their different academic departments.

To use the system, people have to log in. Then students can write, view, and manage their journals, track when they submit items, and generate downloadable PDF reports. Company supervisors have their own space to review and approve, or send back, weekly journals. Academic coordinators can look at what many students are doing, but they cannot change anything. Administrators are responsible for creating accounts, setting up batches, and configuring the platform's settings. Everything users do is stored in the system's database, including journals, approvals, changes, and batch records.

The system is divided into three parts: what the users see, how it works, and where the data is stored. The part the user sees is designed to work well on computers and phones. The part that makes it work includes rules such as creating weekly bundles, sending emails every Sunday, and generating PDF reports. The part that stores data uses a database to store all user information, journals, batches, and system logs. This design ensures the platform is safe, can handle users, and is easy to use for everyone during the OJT term. The project is designed to be flexible and secure. The system is built to meet the needs of students, company supervisors, academic coordinators, and system administrators. The system's architecture is part of the project and plays a major role in ensuring everything works smoothly.

3.2. Details of Technology Used

The purpose of this section is to understand the technologies that developers use and investigate their practical value in different environments.

Microsoft Visual Studio Code – This is the main IDE used by programmers. It gives them a fast and lightweight workspace that supports syntax highlighting, autocomplete, and debug tools, and has extensions for Vue.js, Laravel, React Native, and other languages and frameworks we are using for this project. Also has a built-in terminal and version control.

Laravel – a PHP web application framework that we use to build the server-side logic and core of the application. Laravel is responsible for handling the system's business logic, including automatic journal bundling on Sundays, the email reminder engine, the four-tier review and approval workflow, role-based permissions for the four user types, and the report builder for generating PDFs. Laravel's built-in authentication, scheduling, and database tools fit well with InterTrack's process-heavy ecosystem.

Vue – Vue.js is the JavaScript framework we used to build InternTrack's web view. Vue handles the dynamic rendering of our pages and components, including the student journal page, interactive calendar, supervisor review work page, and coordinator overview page. Vue.js allows InternTrack to update parts of the page without reloading the entire page. Dynamic page updates enable smoother page transitions for users, regardless of their role.

MySQL – is a free database system that helps store and organize information. MySQL stores all data within InternTrack that is created for saving. MySQL's primary purpose is to persistently store all journal entries created throughout the day, week compilations, user and their role privileges, program setup data, edit history, and student internship profile data. MySQL was chosen because its structured table-based format stores information in a relational manner. Tables within MySQL allow for easy querying of dates to create filtered reports. Additionally, MySQL stores Student Information Sheets and SIPP compliance reports. Admins and coordinators can add and retrieve these documents from their dashboards.

React Native – was the primary framework used to develop the mobile application component of InternTrack. With React Native, students and other users can view key system features on their phones, such as adding journal entries, viewing their calendar status, and receiving notifications. React Native supports a single codebase for both Android and iOS platforms, thereby increasing accessibility for interns who may lack consistent access to desktop or laptop computers.

Tailwind CSS – is a utility-first framework that helps design and style InternTrack's user interface. Instead of using pre-made UI components, Tailwind offers utility classes that let developers create custom, consistent, and responsive layouts directly in the markup. The system uses it to style everything from dashboards, forms, and tables to navigation menus, keeping the look clean and professional on both desktop and mobile.

XAMPP – a local server environment used throughout the development and testing stages of InternTrack. XAMPP provides a locally-hosted Apache web server, MySQL database, and PHP interpreter. Effectively, this enables us to run and test the Laravel backend on our own personal computers without having internet access or a hosted server.

Smartphones - are the main devices that students use to access the InternTrack mobile application. Since many interns do not always have acces to their desktop or laptop during their OJT, smartphones offer a convenient, portable way to stay connected. Through the app, students can submit their daily journal entries, check their calendar submissions, and receive notifications when they have overdue or missing submissions on both Android and iOS devices, making it accessible to most students. By supporting smartphones, InternTrack helps students keep up with their OJT documentation and make it easier for them to participate regularly.

Personal Computer - serves as the primary development workstation used by the developers throughout the entire software development lifecycle of InternTrack. Developers use it for writing and testing code, designing system interfaces, managing databases, running local servers, and debugging application behavior. Without a capable personal computer, developers cannot effectively develop, test, and deploy the system.

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